Product, Quotient, and Trigonometric Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.02

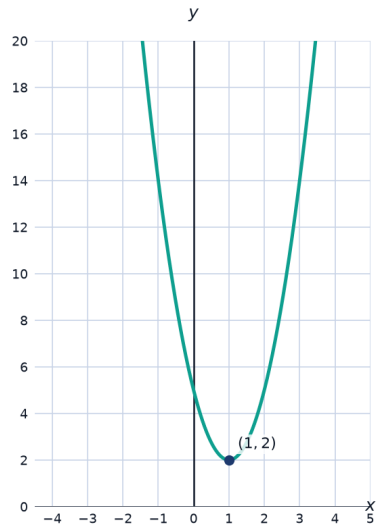
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A differentiable function $f(t)$ is represented by the graph below. What is the total change in $f(t)$ from $t = 2$ to $t = 5$?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = 3x^2 - 6x + 5$. What is the instantaneous rate of change of $f(x)$ at $x = 2$?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A conical tank has height 15 ft and radius 7 ft at the top (radius:height ratio constant as it drains). Water drains at 6 ft^3/min. Find $\frac{dh}{dt}$ (in ft/min) when the water depth is 13 ft.

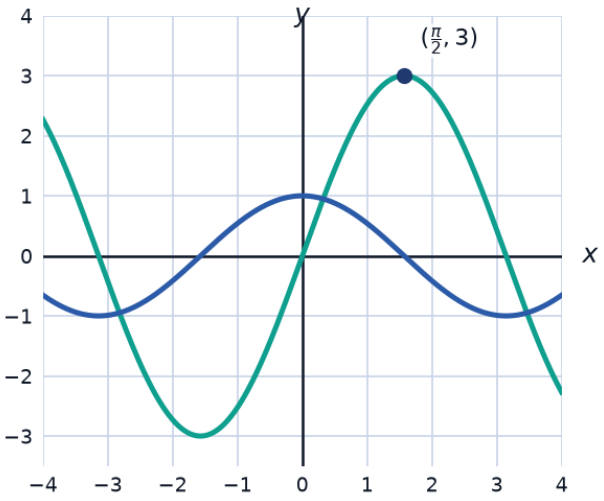
4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Product rule for $\frac{d}{dx}[f(x)g(x)]$:.

© ClipSAT — clipsat.org | All Rights Reserved

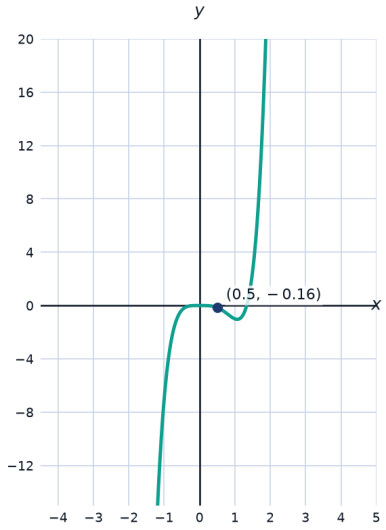
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use the quotient rule to find $f'(x)$ if $f(x) = \frac{x}{4x+2}$.



6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use the product rule to find $f'(x)$ if $f(x) = x^4(3x - 4)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumA 13-foot ladder leans against a vertical wall. The base of the ladder slides away from the wall at a constant rate of 2 ft/s. How fast is the top of the ladder sliding down the wall when the base is 5 feet from the wall?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

© ClipSAT — clipsat.org | All Rights Reserved

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

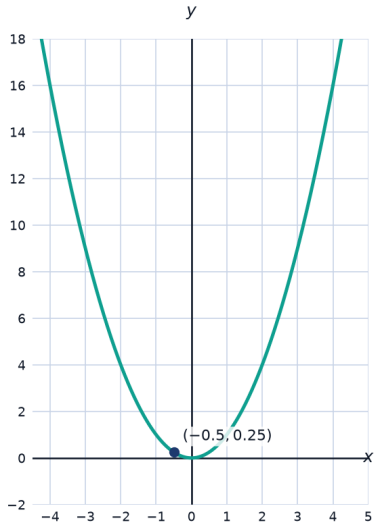
Level 3 · AdvancedLet R be the region bounded by $y = \sqrt{x}$, $y = 2$, and the y-axis. Set up, but do not evaluate, an integral for the volume generated when R is revolved about the vertical line $x = -1$.

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Level 1 · FoundationsExpress the following limit as a definite integral with lower limit $a = 0$, and evaluate it: $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{2}{n}\sqrt{4 - \left(\frac{2i}{n}\right)^2}$

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 3 · AdvancedFind the coordinates of the point on the parabola $y = x^2$ that is closest to the point $(0, 3)$. Prove that this distance is a global minimum.



11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 2 · MediumFind the equations of all horizontal and vertical tangent lines to the ellipse $4x^2 + 9y^2 = 36$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 1 · FoundationsA particle moves along a line with position $s(t) = 2t^3 - 15t^2 + 24t + 10$. Determine the open intervals on $t > 0$ during which the particle is speeding up.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumFind the particular solution to the differential equation $\frac{dy}{dx} = \frac{x}{y}$ with initial condition $y(0) = -4$. State the domain of the solution.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · AdvancedA population of bacteria grows at a rate proportional to its size: $\frac{dP}{dt} = kP$. At $t = 2$ hours, the population is 400. At $t = 6$ hours, the population is 1600. What was the initial population $P(0)$, and what is the doubling time?

© ClipSAT — clipsat.org | All Rights Reserved

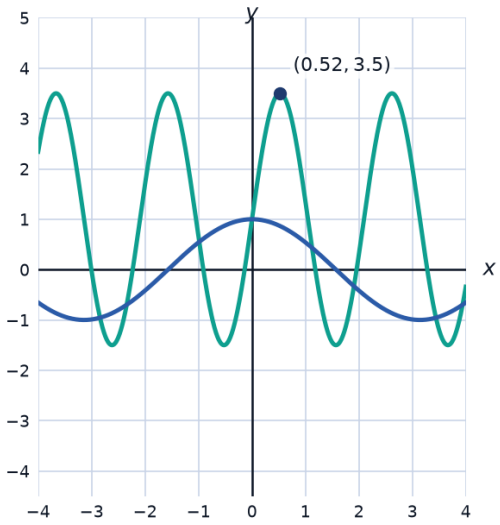
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 The Chain Rule**

Level 1 · FoundationsWhich of the following functions is a solution to the differential equation $y'' + 4y = 0$? (A) $y = e^{2x}$ (B) $y = \sin(2x) - 3\cos(2x)$ (C) $y = 4\cos(4x)$ (D) $y = x^2 - 4$

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Level 2 · MediumFind the value of constant k that makes $f(x) = \{kx^2 + 1, x \le 2; 3x - 5, x > 2$ continuous at $x = 2$.



17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

[Unit 1 · Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

[Unit 3 · Differentiation — Composite, Implicit, and Inverse Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

[Unit 4 · Contextual Applications of Differentiation · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

[Unit 5 · Analytical Applications of Differentiation · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the second derivative $f''(x)$ of $f(x) = \sin x$.

a. $-\sin x$

b. $\sin x$

c. $-\cos x$

d. $\cos x$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If position is $s(t) = t^4 - 2t^3 + 5$, what is the acceleration $a(t)$ at $t = 2$?

a. 24

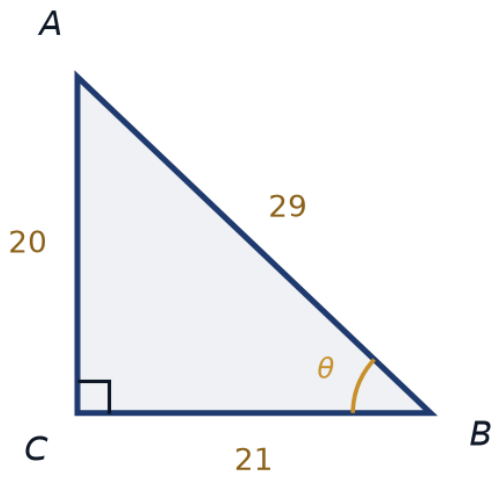
b. 36

c. 12

d. 48

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the 50th derivative of $f(x) = \cos x$?



a. $-\cos x$

b. $\cos x$

c. $\sin x$

d. $-\sin x$

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the chain rule to find $f'(x)$ if $f(x) = (-3x - 3)^4$.

a. $4(-3x - 3)^3$

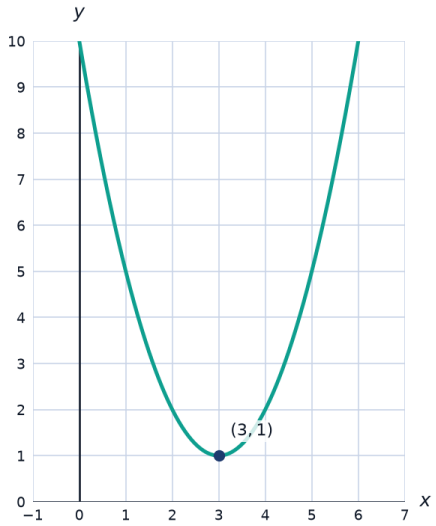
b. $-12(-3x - 3)^4$

c. $-12(-3x - 3)^3$

d. $-3(-3x - 3)^3$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.



a. x = 3

b. x = -3 and x = 3

c. x = 1 and x = 5

d. x = 1

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. 4

d. 8

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An AP Calculus BC FRQ models a population $P(t)$ satisfying the logistic differential equation $\frac{dP}{dt} = \frac{3}{600}P(600 - P)$. At what population size P is the population growing most rapidly?

a. $P = 300$

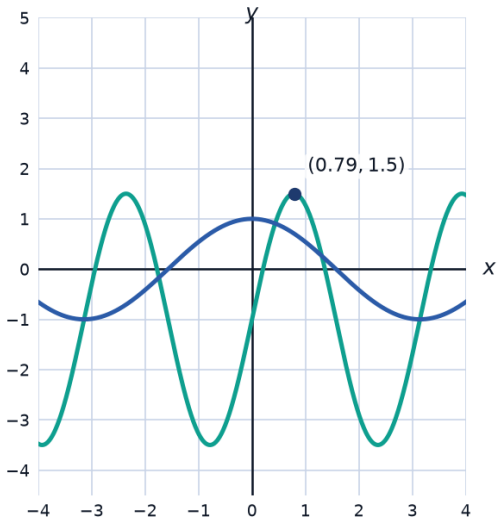
b. $P = 600$

c. $P = 150$

d. $P = 0$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

On an AP Calculus BC Free-Response Question, a particle moves along a curve with velocity vector $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. What is the speed of the particle at time $t = 1$?



a. $\sqrt{13}$

b. 5

c. $\sqrt{5}$

d. 13

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a Taylor series BC FRQ, the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ is approximated by its first 2 terms S_2. According to the Alternating Series Error Bound, what is the maximum possible error $|S - S_2|$?

a. $\frac{1}{3^3} = \frac{1}{27}$

b. $\frac{1}{2^3} = \frac{1}{8}$

c. $\frac{1}{3^2} = \frac{1}{9}$

d. $\frac{1}{2 \cdot 3^3}$

© ClipSAT — clipsat.org | All Rights Reserved

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Let R be the region bounded by the graph of $y = \sin(\pi x)$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the x-axis. The radius of a cross-section of the solid perpendicular to the x-axis is given by $r(x) = k\sin(\pi x)$ for some positive constant k. If the volume of the solid is $\frac{\pi}{2}$ cubic units, determine the value of k.

a. 1/2

b. 1

c. 2

d. 4

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The base of a solid is the region under $y = 3\sqrt{x}$ on $[0, 4]$. Cross-sections perpendicular to the x-axis are squares with side length y. Find the volume of the solid.

a. 72

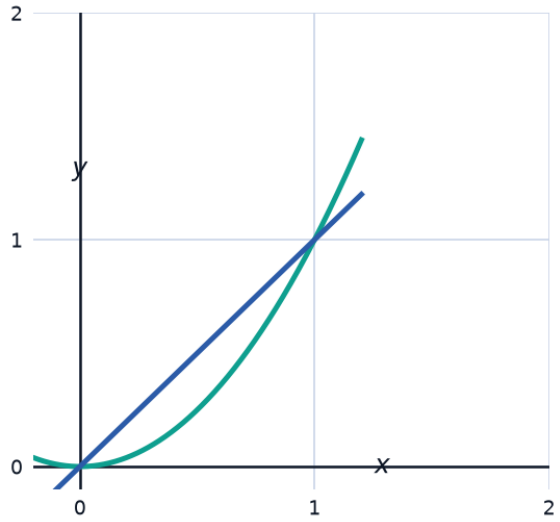
b. 74

c. 144

d. 70

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Area between $y = x^2$ and $y = x$ from 0 to 1 is:



a. $\frac{1}{3}$

b. $\frac{1}{6}$

c. $\frac{1}{2}$

d. $\frac{2}{3}$

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A population of bacteria, denoted by $P(t)$, grows according to the differential equation $\frac{dP}{dt} = kP$, where k is a constant representing the growth rate. It is observed that the population triples every 4 hours. If the initial population at $t = 0$ is $P_0 = 1000$, determine the population after 12 hours.

a. 10000

b. 27000

c. 81000

d. 9000

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A particle's position in the first quadrant is described by a function $y(t)$, where t is time in seconds. The rate of change of its y-position with respect to time is governed by the differential equation $\frac{dy}{dt} = k\sin(t)$ for some positive constant k. The particle starts at the origin $(0, 0)$ at $t = 0$. At $t = \frac{\pi}{2}$ seconds, its y-position is $y(\frac{\pi}{2}) = 5$. \\ \\ Consider a second particle whose y-position $z(t)$ is described by the differential equation $\frac{dz}{dt} = k\cos(t)$, also starting at the origin at $t = 0$. If the first particle's y-position at $t = \frac{\pi}{4}$ is $y(\frac{\pi}{4}) = \frac{5}{\sqrt{2}}$, what is the value of k?

a. $\frac{10}{\pi}$

b. 5

c. $\frac{5\sqrt{2}}{\pi}$

d. $\frac{5}{2}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A rectangular garden has a length that is twice its width. If the perimeter of the garden is 36 meters, what is the width of the garden?

a. 6 meters

b. 9 meters

c. 12 meters

d. 18 meters

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

$f(x) = x^3 - 3x$. Inflection point at:

a. $x = -1$

b. $x = 3$

c. $x = 0$

d. $x = 1$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $\frac{dy}{dx} = \frac{x}{y}$ given $y(0) = 6$ (assume $y > 0$).

a. $y^2 + x^2 = 36$

b. y - x = 6

c. $y^2 - x^2 = 36$

d. $y^2 - x^2 = 38$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

A particle moves with velocity $v(t) = 3t^2$. Displacement from $t = 0$ to $t = 2$ is:

a. 8

b. 24

c. 12

d. 16

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Use linearization (the tangent line) of $f(x) = 1\sqrt{x}$ at $x = 2$ to approximate $f(2 - \frac{1}{10})$.

a. $-1 + \frac{39\sqrt{2}}{40}$

b. $\sqrt{2}$

c. $1 + \frac{39\sqrt{2}}{40}$

d. $\frac{39\sqrt{2}}{40}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

For $f(x) = (x - 5)(x - 9)$, determine the intervals where f is increasing and decreasing.

a. decreasing everywhere

b. increasing everywhere

c. increasing on $(-\infty, 7)$, decreasing on $(7, \infty)$

d. decreasing on $(-\infty, 7)$, increasing on $(7, \infty)$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

A population satisfies $\frac{dP}{dt} = \frac{1}{50}P$ with $P(0) = 390$. Find $P(100)$.

a. $P = 390 \cdot 2$

b. $P = 390e^{-2}$

c. $P = 390e^{2}$

d. $P = 390e^{2/100}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

If $f(x) = x^3 + 2x^2 - 6x$, is the graph of f concave up or concave down at $x = 5$?

a. cannot be determined

b. concave up

c. concave down

d. neither

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the area of the region above the x-axis bounded by $y = -(x - 4)(x - 11)$.

a. $\frac{361}{6}$

b. $\frac{343}{6}$

c. $\frac{325}{6}$

d. $\frac{343}{3}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Area Between Curves

Find the area of the region enclosed by $y = x^2$ and $y = 7x$.

a. $\frac{355}{6}$

b. $\frac{343}{6}$

c. $\frac{331}{6}$

d. 49

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

Solve the initial value differential equation $\frac{dy}{dx} = \frac{4x}{y}$ with condition $y(0) = 3$ (for $y > 0$).

a. $y = \sqrt{4x^2 + 9}$

b. $y = \sqrt{4x^2 - 9}$

c. $y = 4x^2 + 3$

d. $y = \sqrt{8x^2 + 9}$

© ClipSAT — clipsat.org | All Rights Reserved